

Research Project

Windows BOF- Collections

Offensive security
research and capability
testing without the C2
overhead.

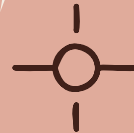


whoami



Name

CJ Enverzo



Area of Interest

Red Team Operations and Development



Blog and Project

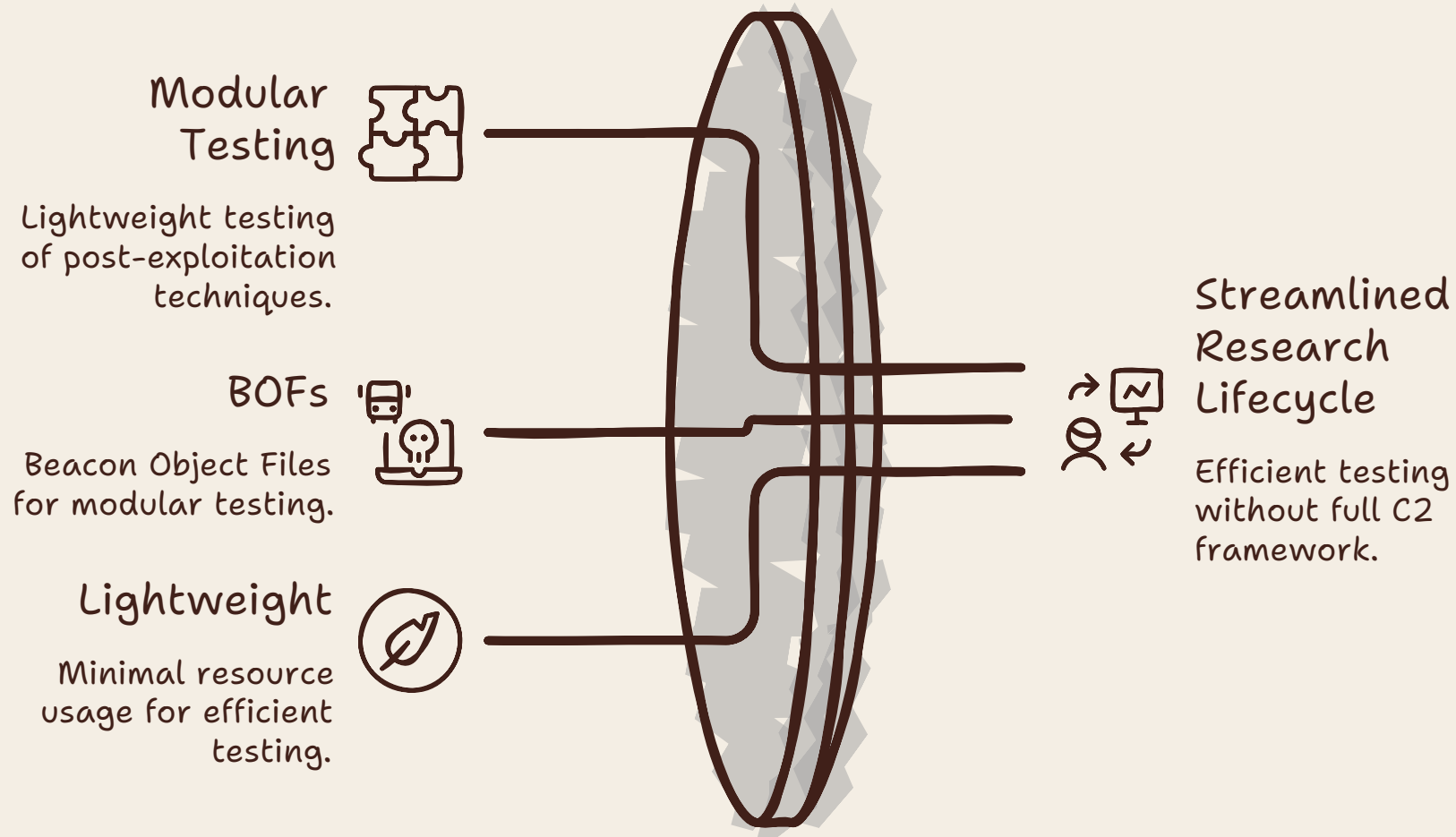
mooolight.github.io / github.com/mooolight



Certificate and Trainings

ZeroPointSecurity Certified Red Team Operator [CRTO] (ongoing), Red Team Infrastructure Developer Certificate [CRT-ID], Windows System, Kernel Programming and Internals, ReTrain Canada/Manpower Penetration Testing certificate, CompTIA Security+, CompTIA A+, Certified CyberDefender [CCD] Pathway, Elearnsecurity Jr. Penetration Testing [eJPT], TryHackMe Red Team Path, TryHackMe SOC L1

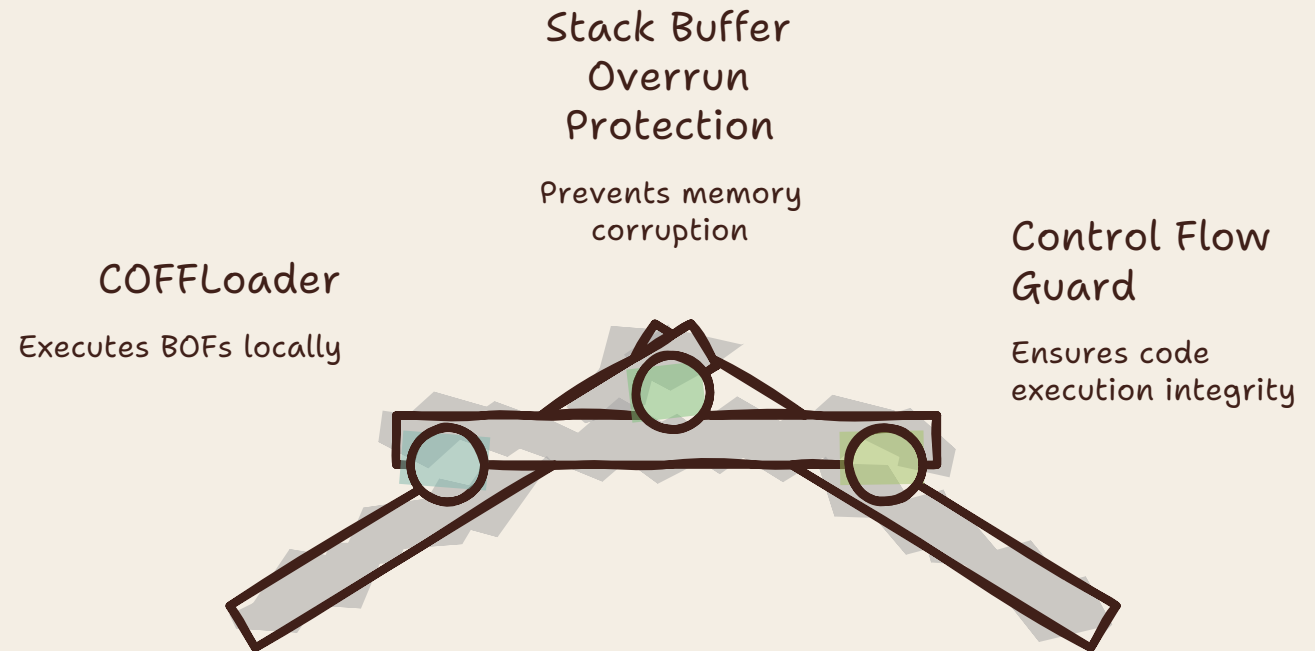
Modular Testing Efficiency



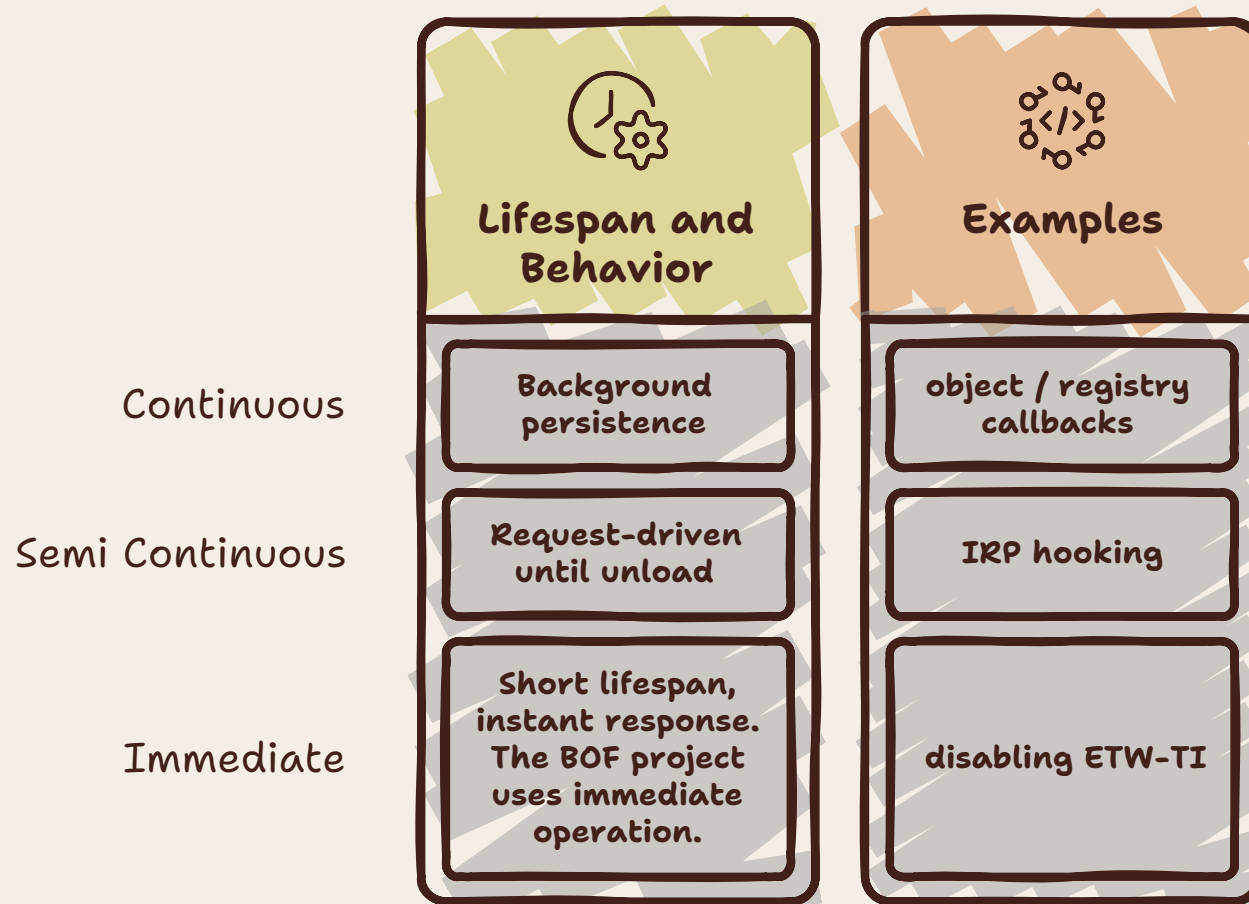
Testing Environment and Tools

Leveraging TrustedSec's COFFLoader for local BOF execution.

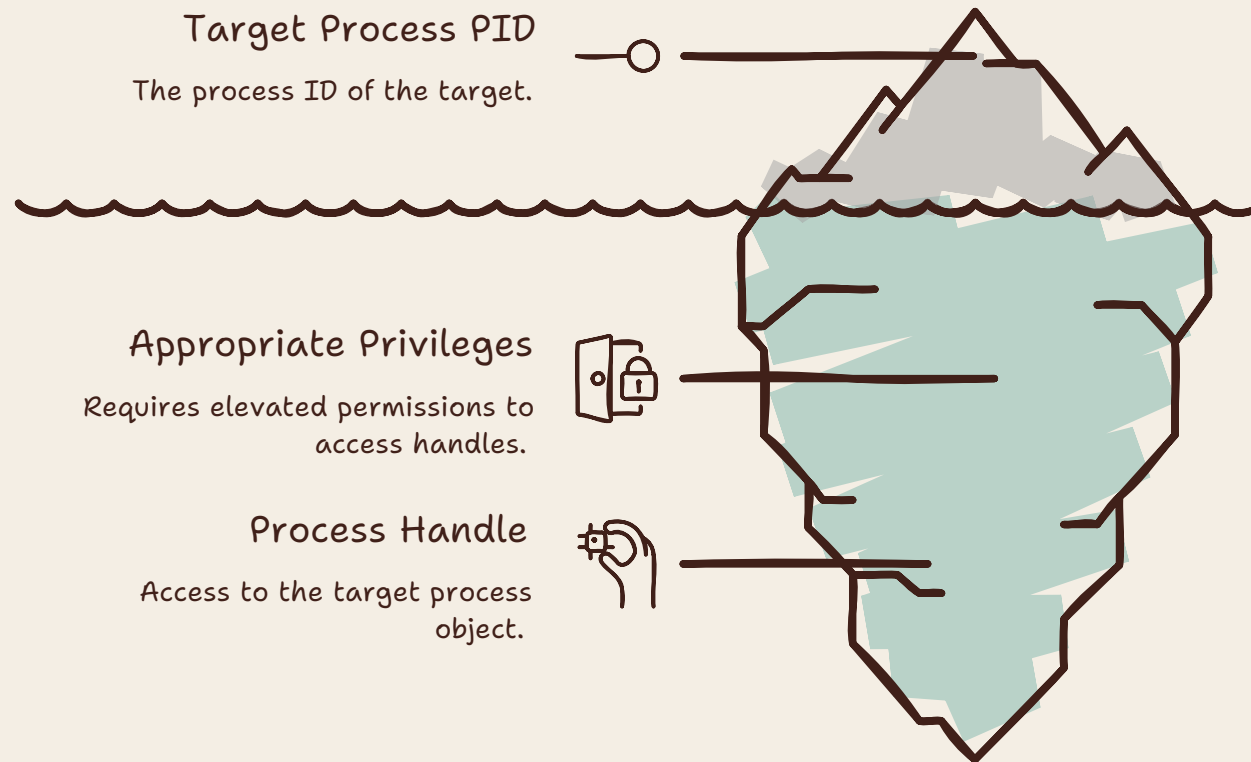
- Tool: COFFLoader from TrustedSec (<https://github.com/trusted-sec/COFFLoader>)
- Protections: Compiler's Stack Buffer Overrun protection ('/GS-') and Control Flow Guard (CFG) enabled during testing.
- Control Flow Guard



Operational Architecture: BOF Execution

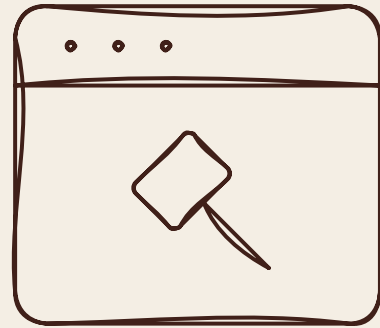


Procenum: Unveiling Target Process PIDs



Credential Access: Browser Exploitation

Extracting sensitive data from common web browsers.



100%
edgelord



Scans msedge.exe's
process

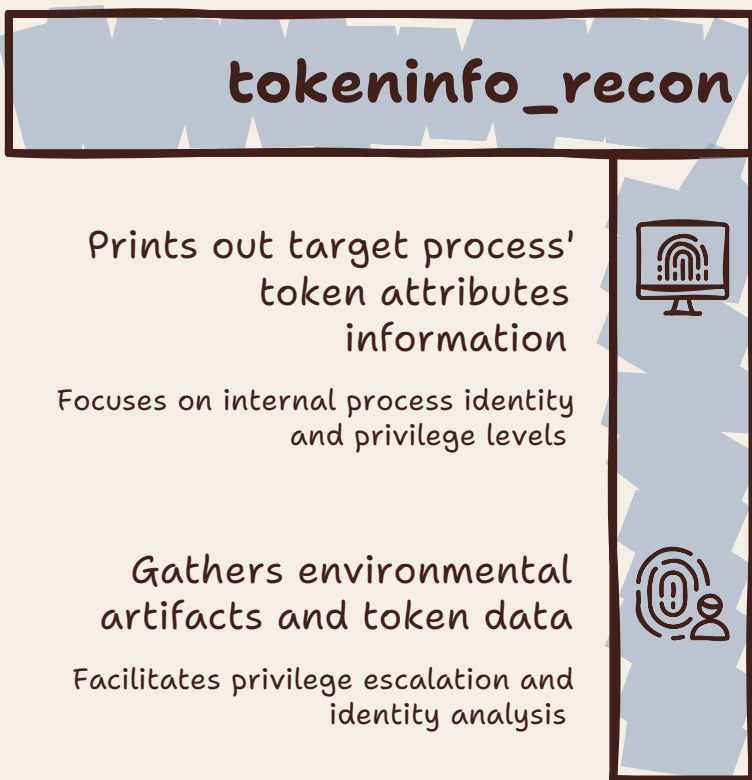


100%
cookiecrunch



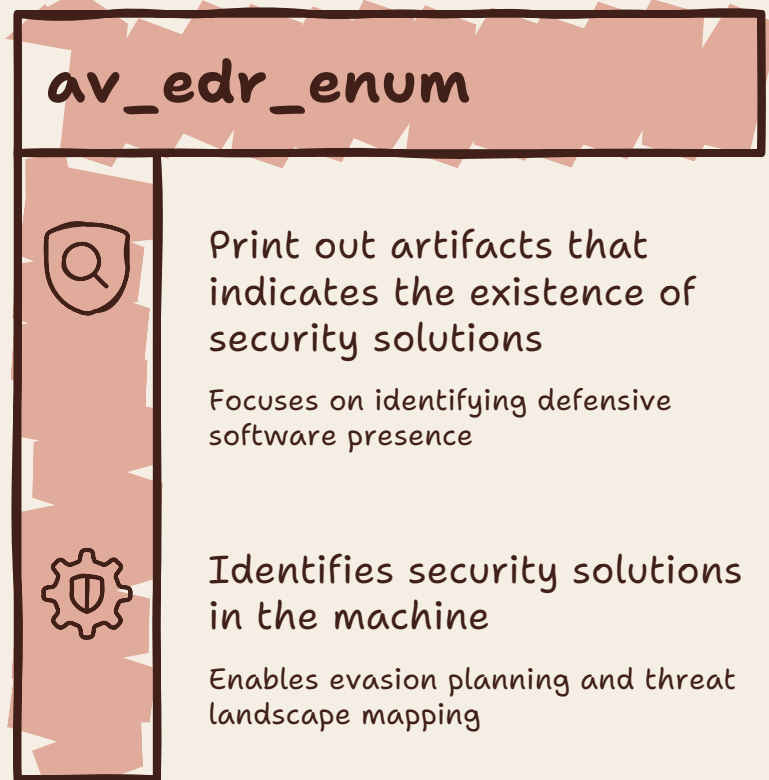
Extracts cookies for
common browsers

Which reconnaissance tool aligns with your current operational objective?



Primary Function

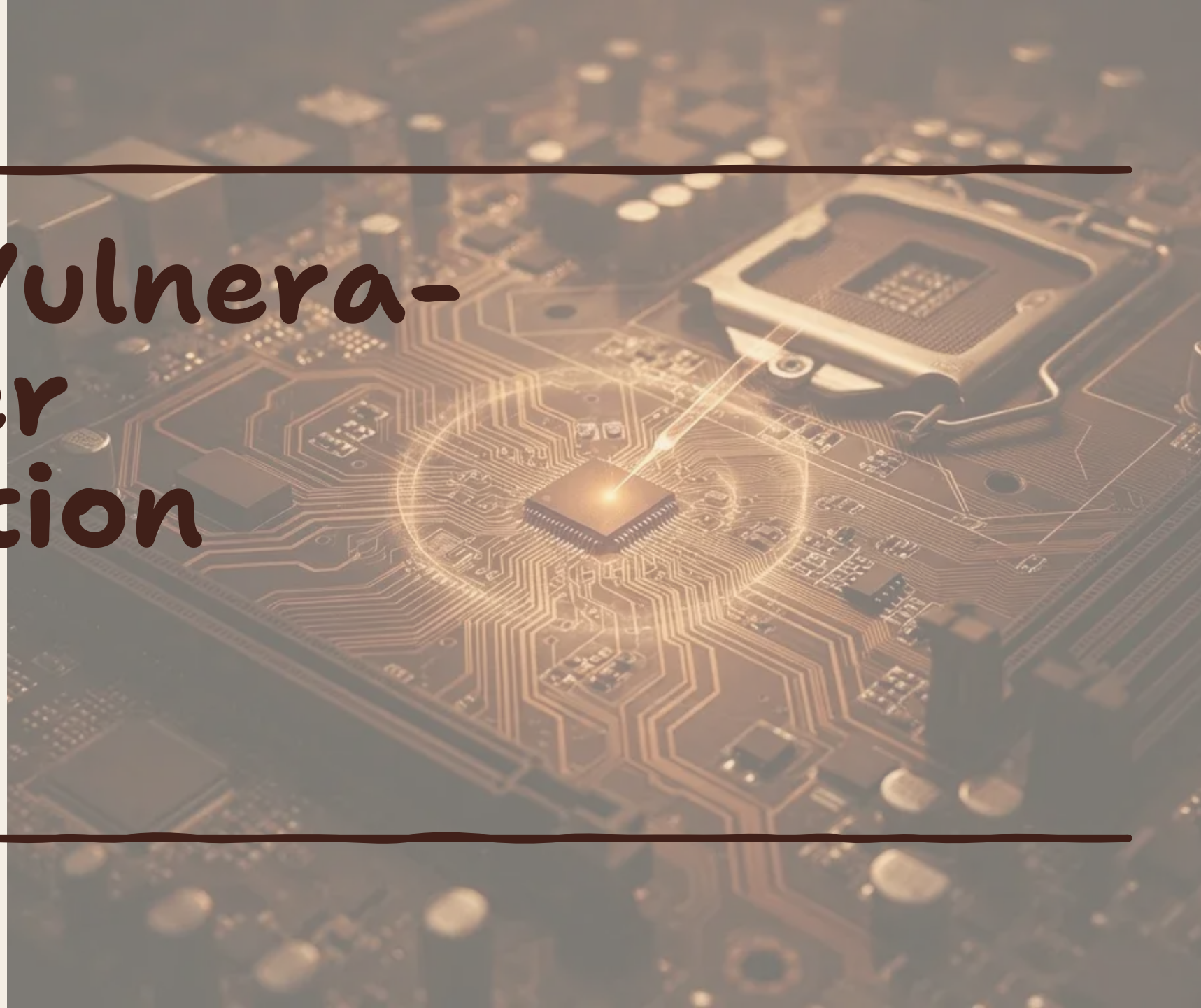
Strategic Utility



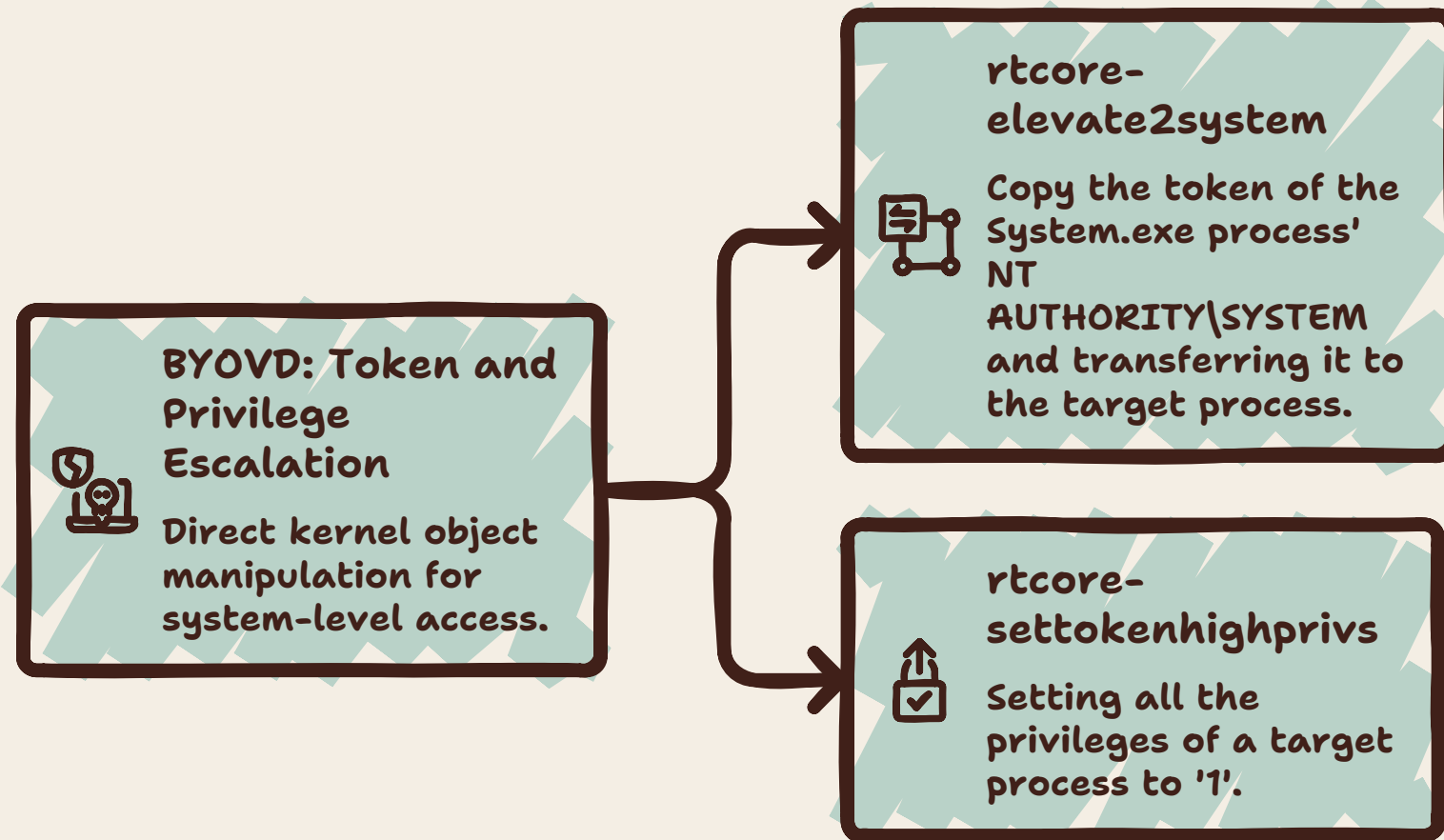
BYOVD

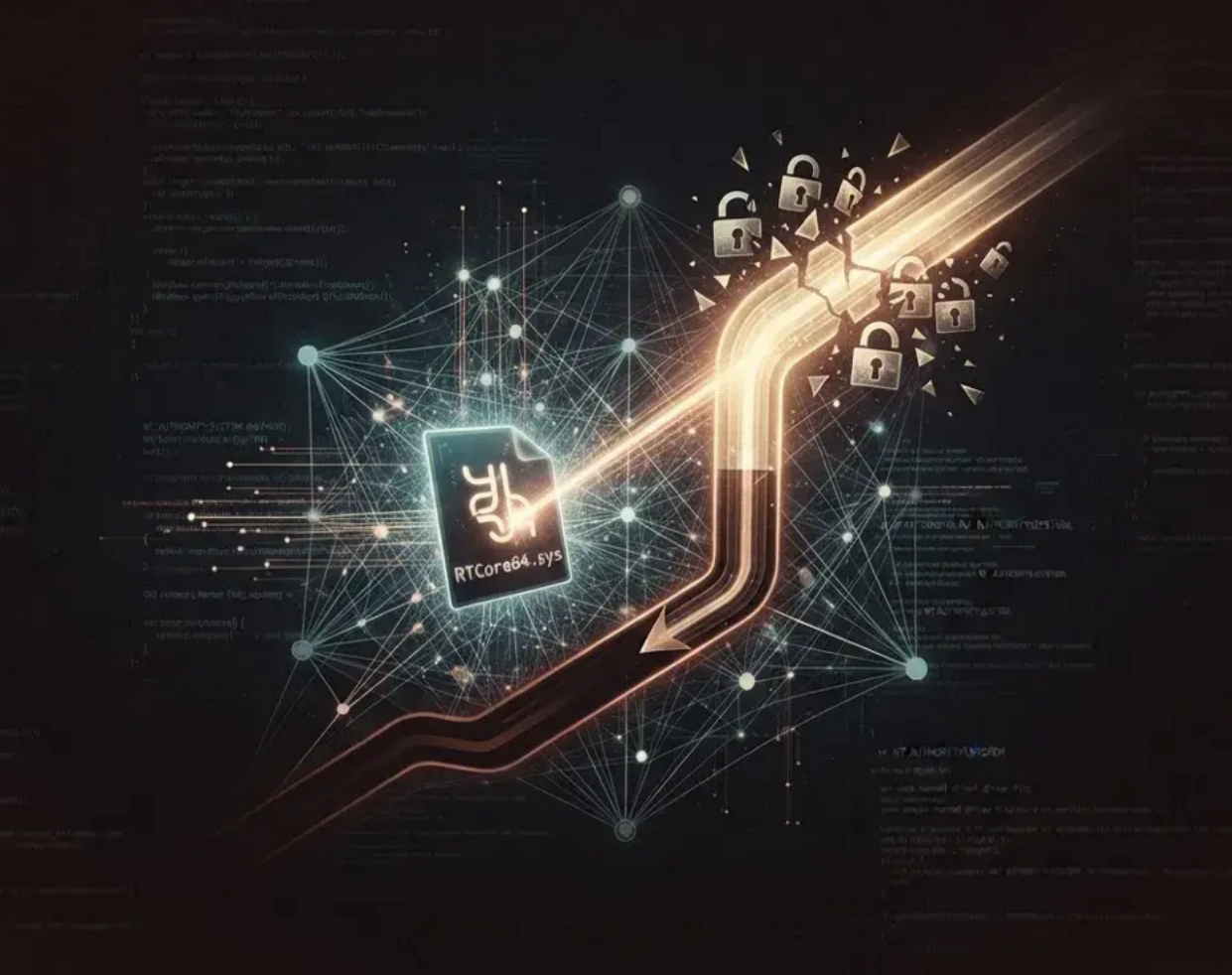
BYOVD: Vulnerable Driver Exploitation

Leveraging
RTCore64.sys for
kernel-level
manipulation.



BYOVD: Token and Privilege Escalation Process





BYOVD: Integrity and Restrictions

Bypassing SID restrictions and elevating integrity levels.

`rtcore-setintegritylevel`

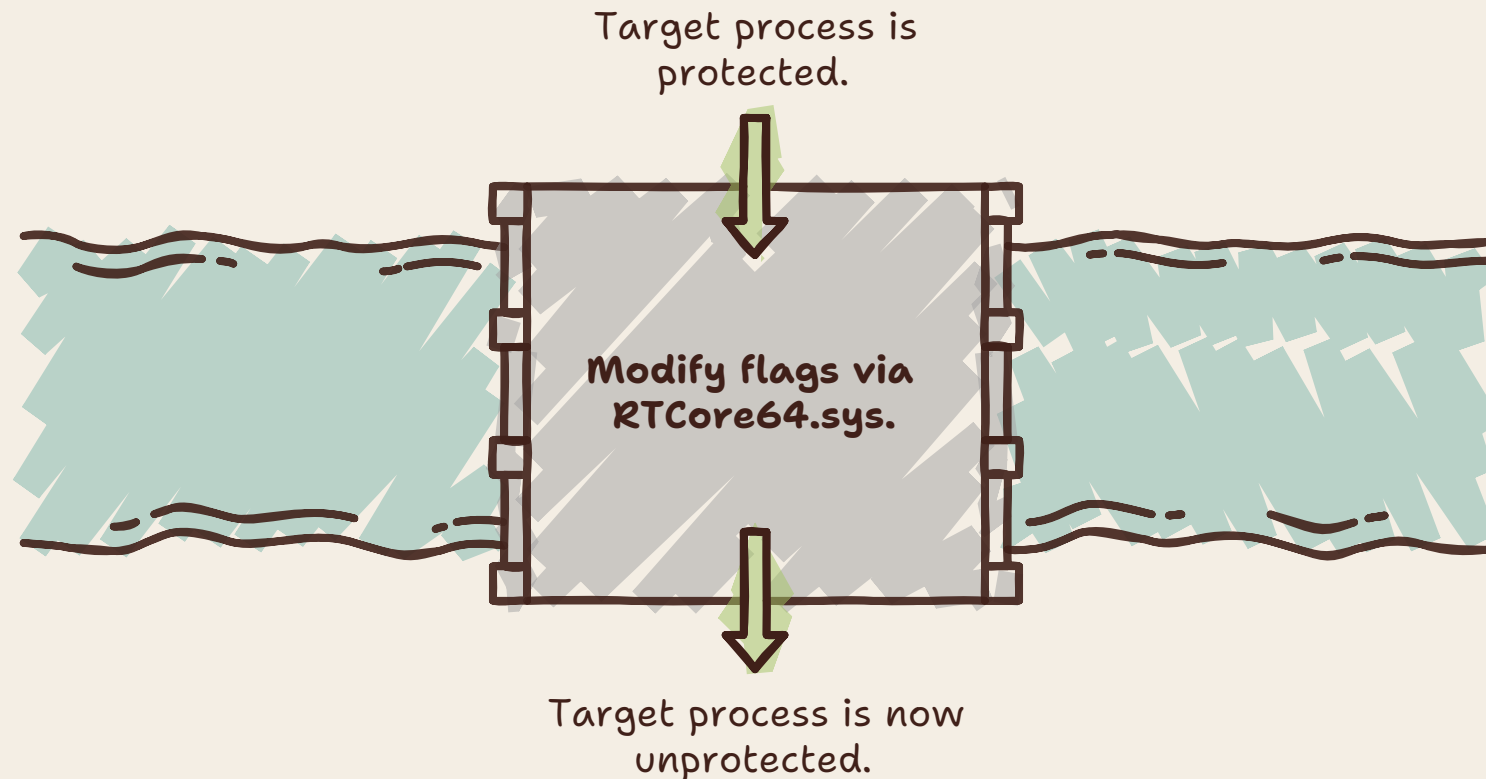
Elevate the integrity of the target process to NT AUTHORITY\SYSTEM.

`rtcore-unrestricttoken`

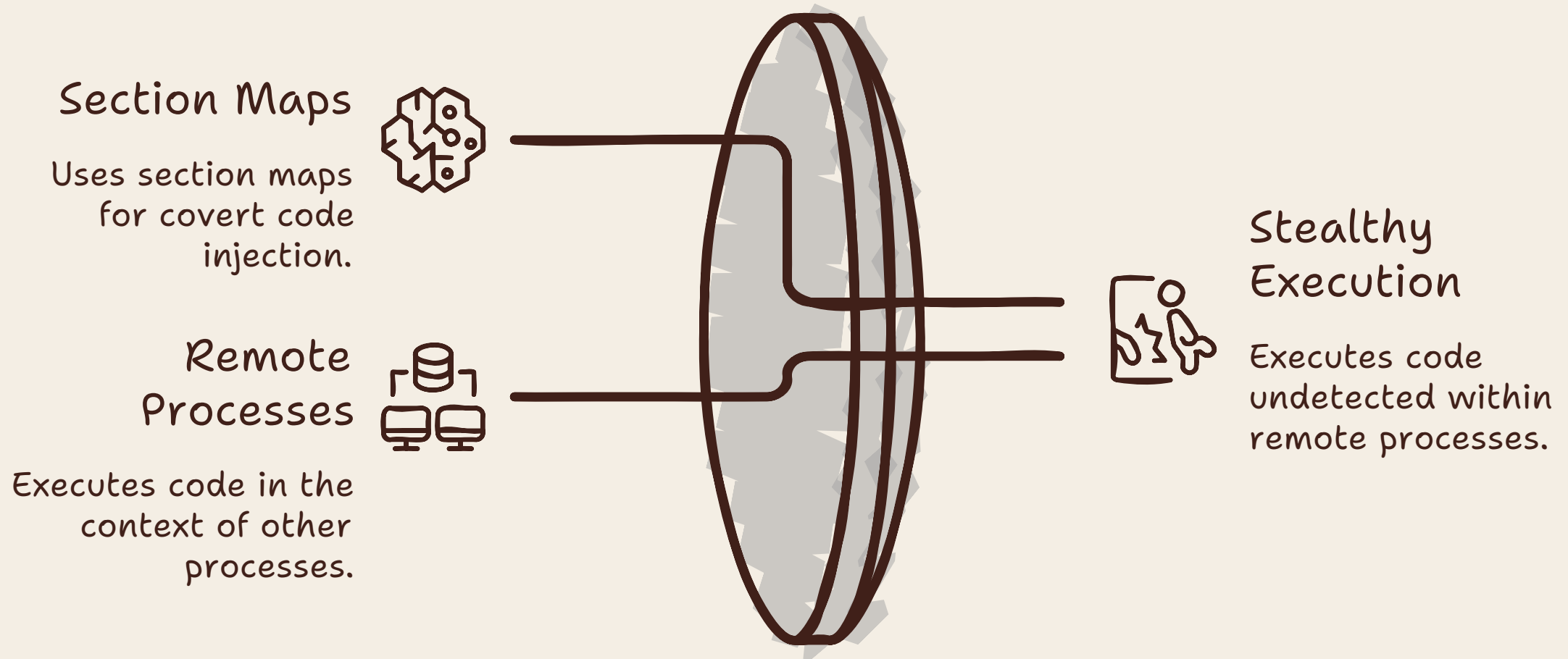
Allows access of the target process to any objects in the system by exploiting RTCore64.sys.

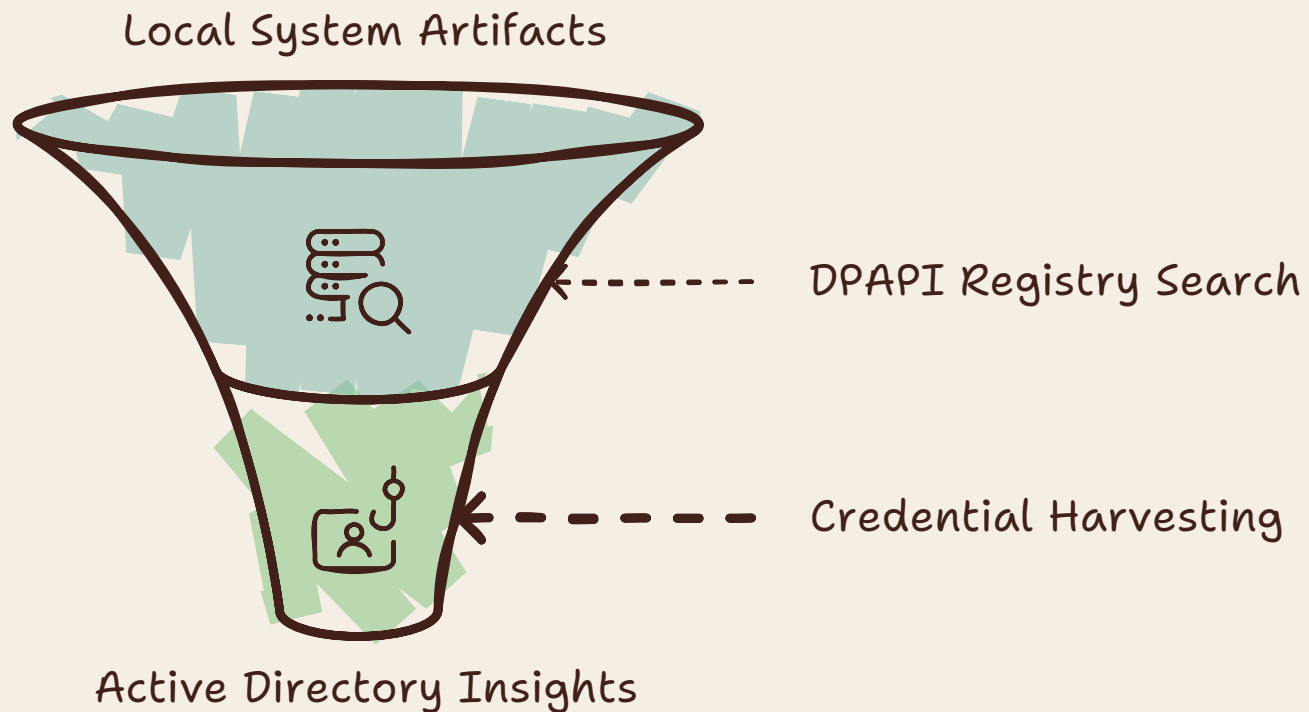
BYOVD: Process Protection Bypasses

rtcore-flipprocprotection: Flip the protection of the target process to Fully Protected Process or Process Protected Light.



Process Injection Techniques



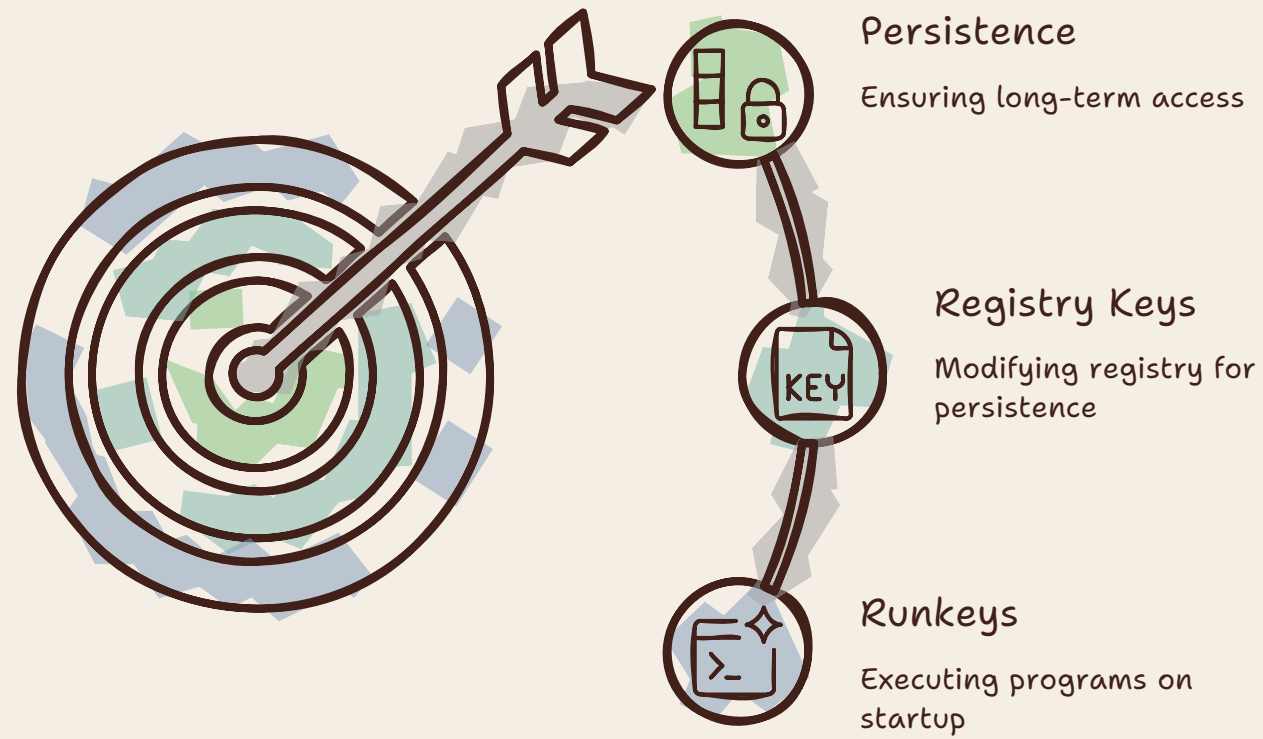


Active Directory Discovery

dpapi-regsearch: Find artifacts related to DPAPI blob in the Windows Registry to facilitate credential harvesting.

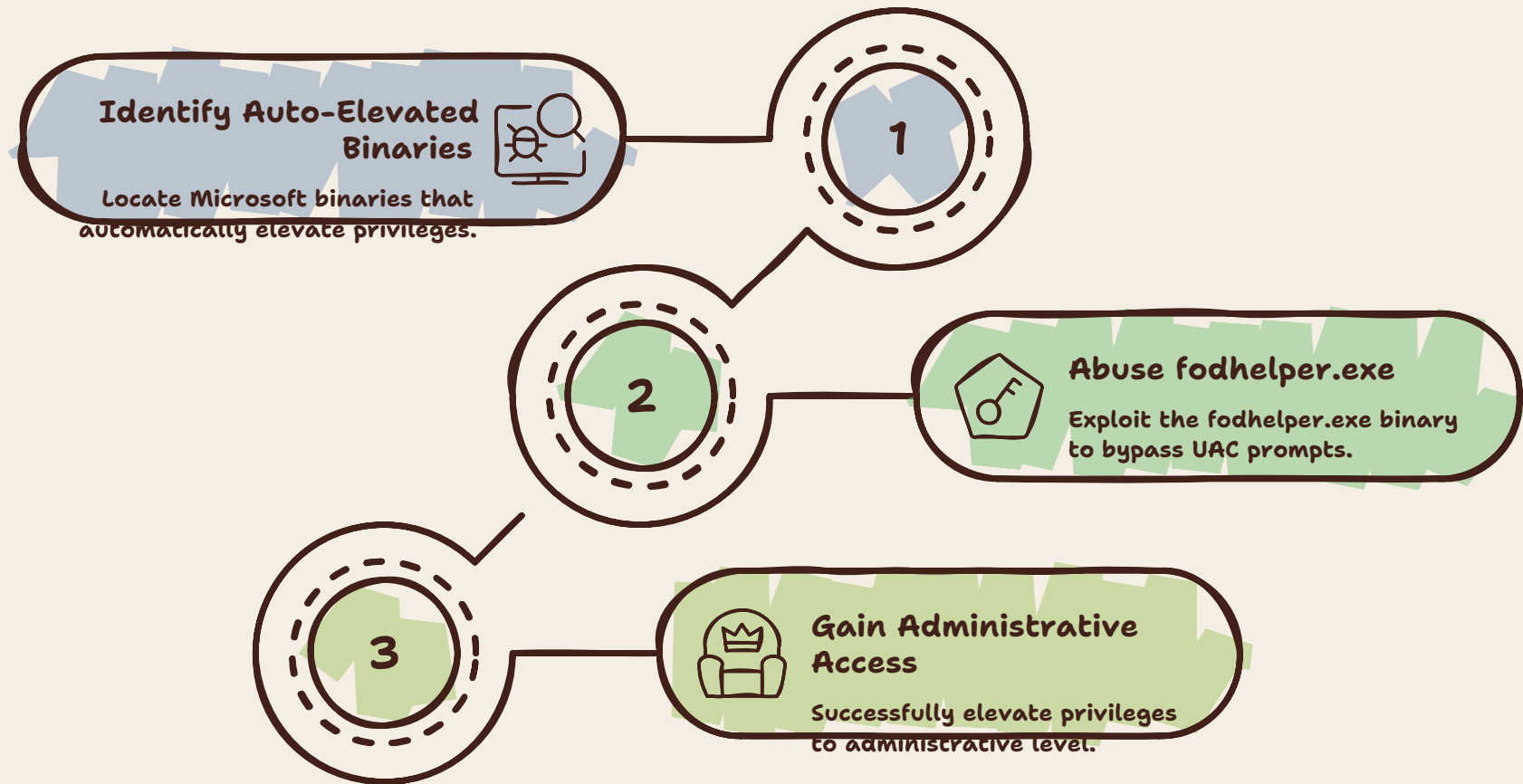
Persistence Mechanisms

`pers-runkeys`: Executes the runkeys persistence technique to ensure access survives reboots.



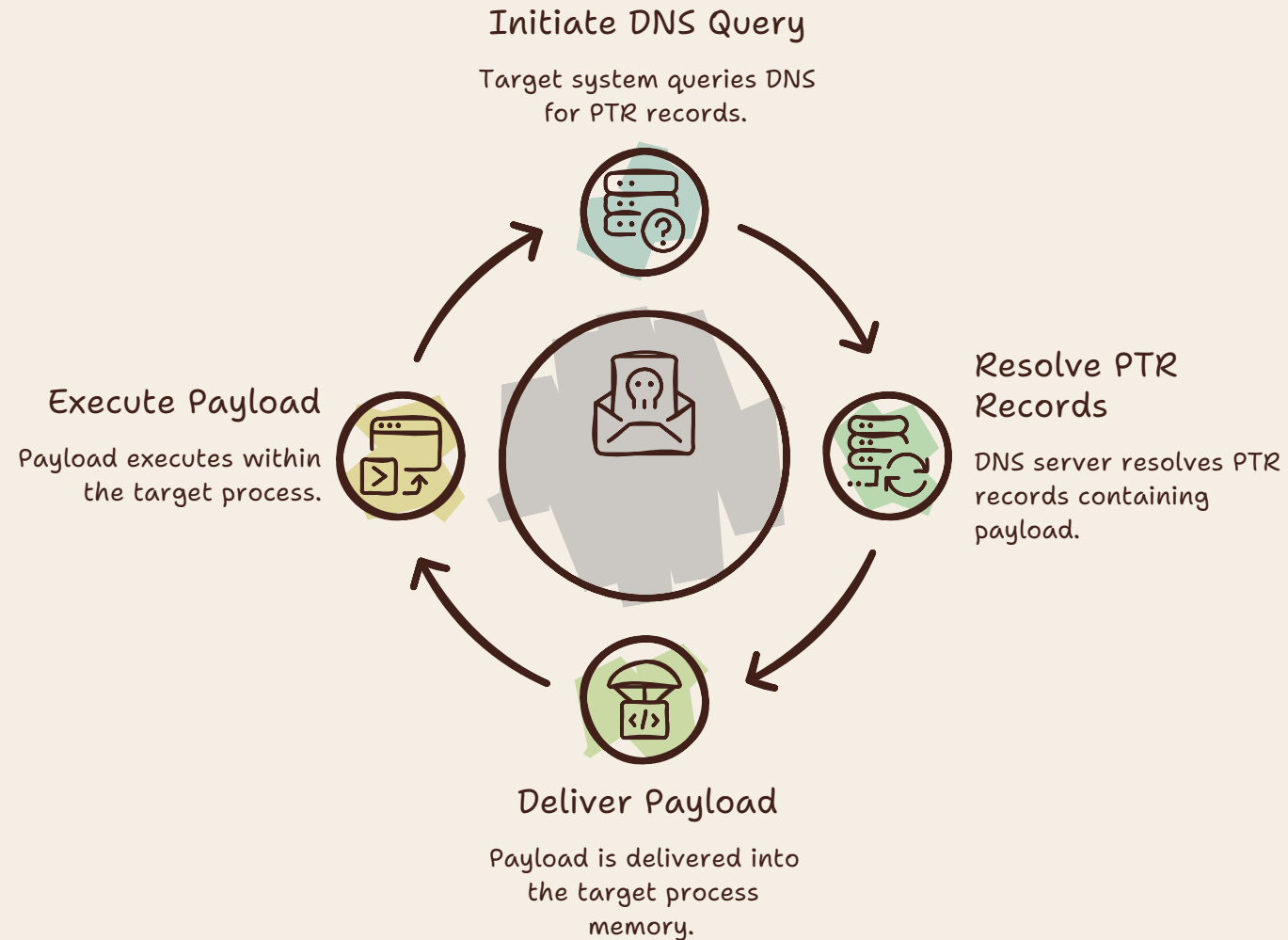
Privilege Escalation: UAC Bypass

Abusing auto-elevated binaries for administrative access.

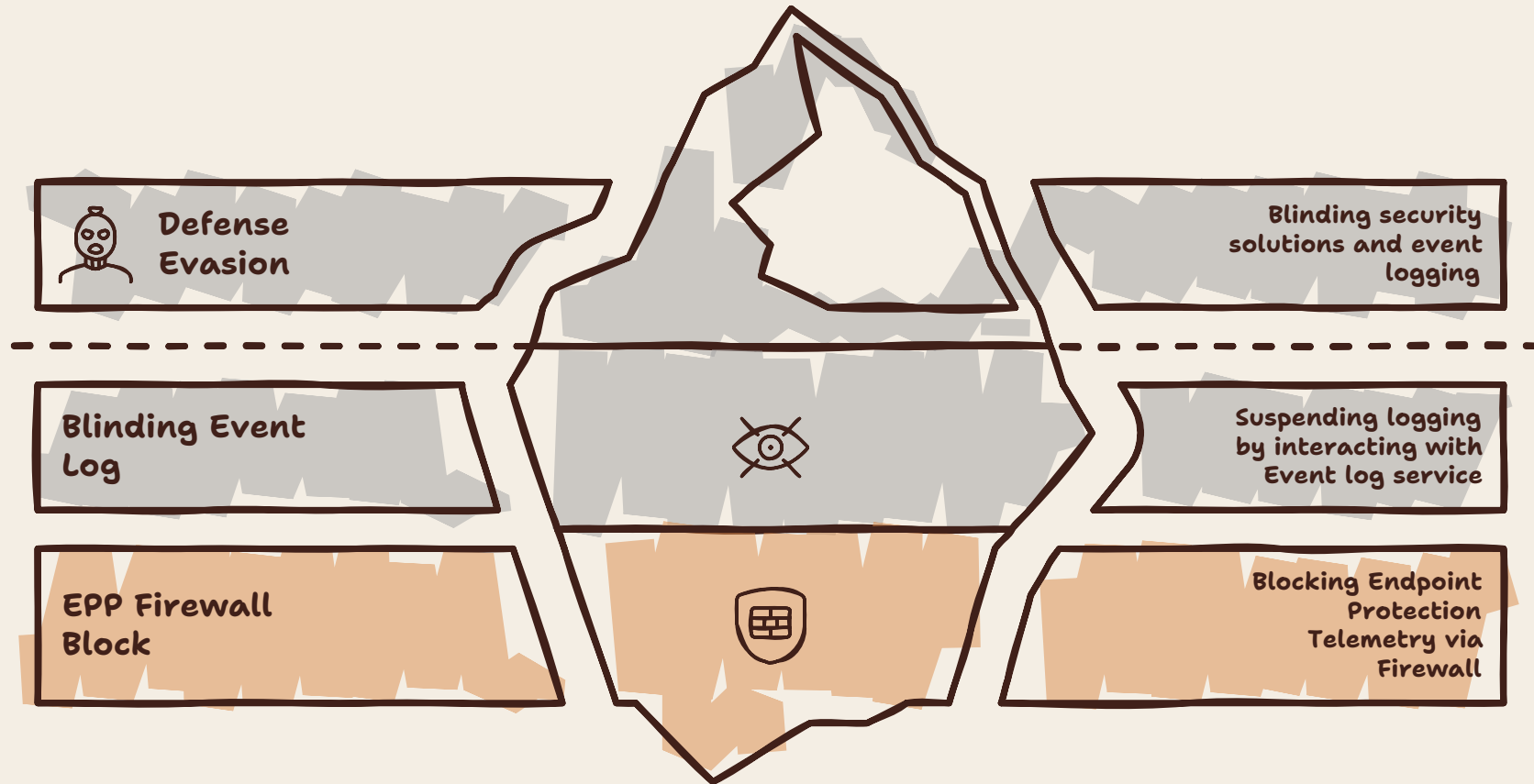


Malware Delivery via DNS

Payload delivery through non-standard channels.

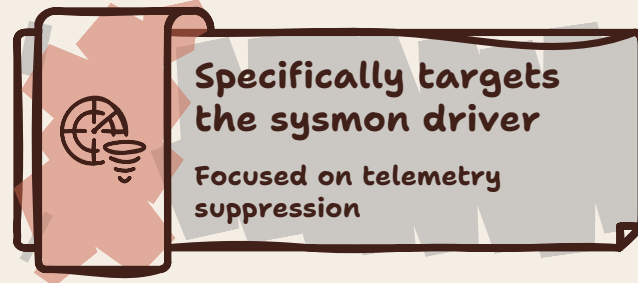
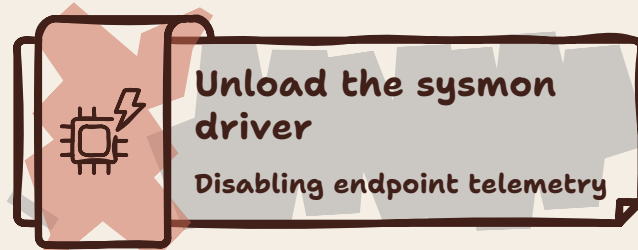


Blinding Security Solutions: A Defense Evasion Tactic



Which defense impairment strategy should be prioritized?

SysmonUnload

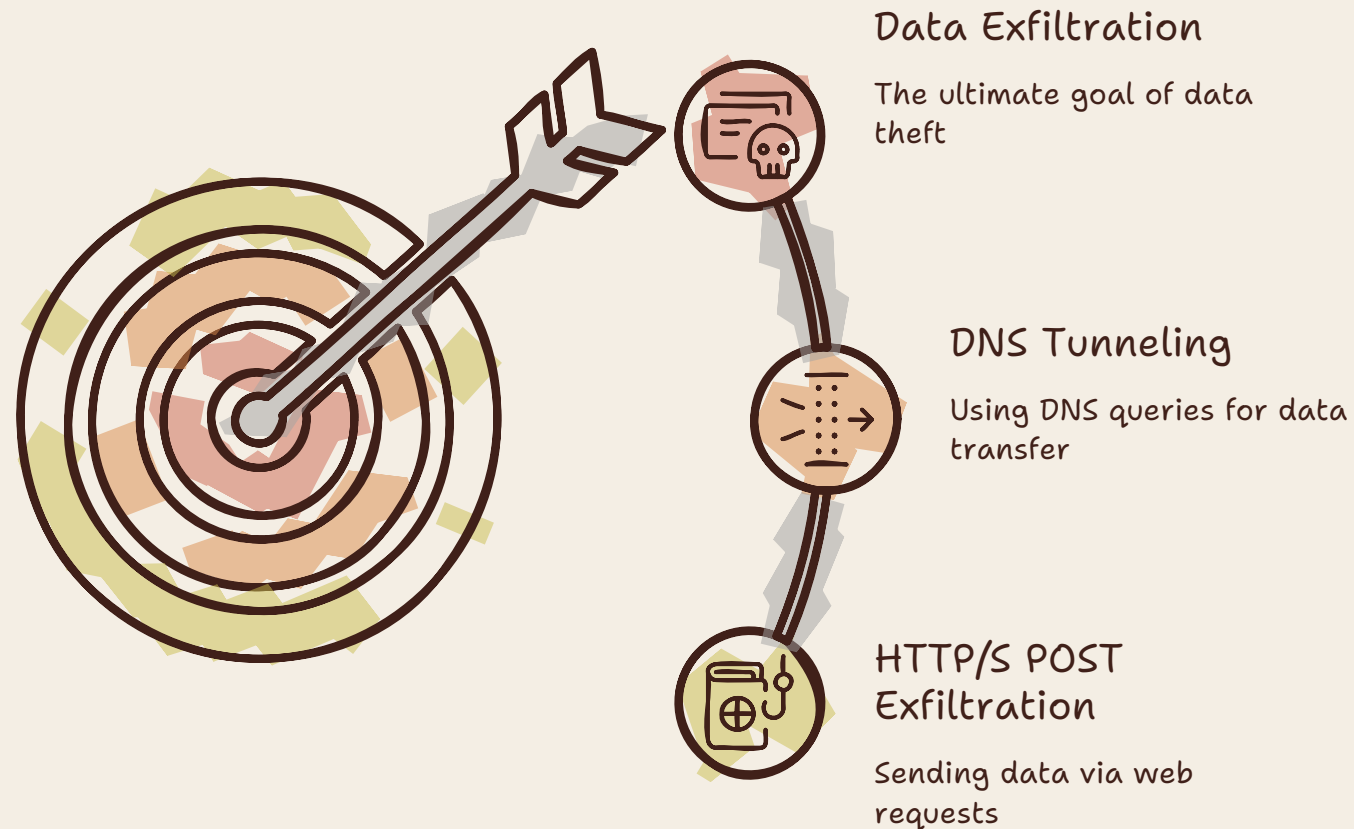


windef-disable



Data Exfiltration (Placeholder)

Placeholder for upcoming data exfiltration research and BOF implementations.



Impact: Ransomware Simulation

ransomware-sim-bof: Executes a ransomware simulation on a proof of concept file with the most restricted access control.

Demonstrate Impact

Show the final stage of the attack lifecycle.

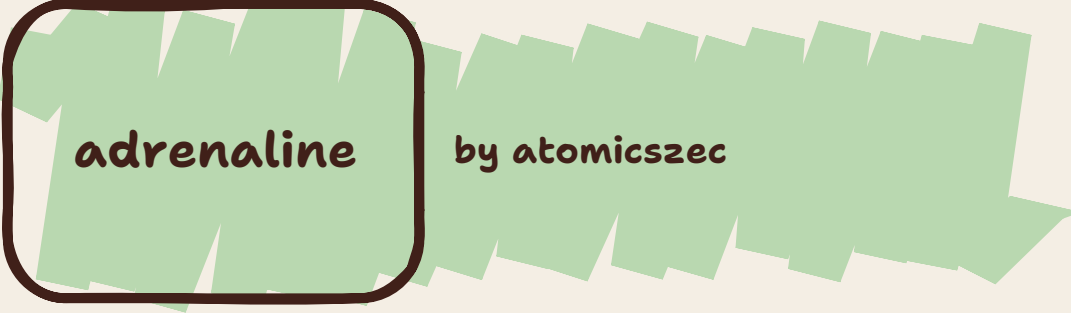
Execute Simulation

Run the ransomware simulation on the restricted file.



Research References

Direct project references
and inspirations from the
community.



adrenaline

by atomicszec



atomic-bofs

by Rasta-Mouse